9 **Ans: B**

t = 0 to P (Initial Fall)

Because the slope is positive but decreasing, the acceleration must start at a high positive value (9.81 ms^{-2}) and decrease toward zero.

P to Q (First Terminal Velocity)

Zero slope means zero acceleration. The parachutist has reached first terminal velocity where air resistance exactly balances weight.

Q to R (Parachute Opens)

A steep negative slope means a large negative acceleration (deceleration).

Comparing the magnitude of the initial acceleration (at t = 0) with the magnitude of the deceleration (at t = Q):

At t = 0, the only force acting is gravity. At release, the air resistance is initially zero, so the acceleration is g

At t = Q, the parachute opens. The air resistance is greater than the person's weight. Therefore, the deceleration spike at Q must be much larger (extend further negatively) than the initial positive acceleration starting point.

Graph B correctly shows a large negative spike.

R to S (Second Terminal Velocity)

Acceleration returns to zero.